

Tide retrospective: grade recovery the semi-quasi-homogeneous endpoint

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Abstract

The last two stages of the scoping consult’s plan for the germbij note’s constructive class (c), and its designated endpoint. Stage C is covariance injectivity: a continuous observable with vanishing self-covariance under a positive reference density is almost everywhere its mean — the L^2 argument $\int (Q - m)^2 e^{-P} = Z \cdot \text{Cov}_P(Q, Q) = 0$ — hence constant by continuity against the open-positive volume, and a vanishing value at the origin kills the constant; finite combinations of nonconstant monomials need only the covariances against their own support. Stage D composes this with the one-grade difference limit: when the normalized moment differences of two families vanish at the grade rate for every support monomial while the difference limits equal the negative covariances, uniqueness of limits forces the grade difference to vanish identically. With the exact moment-ratio recovery for principal quasi-homogeneous models, the consult’s endpoint is realized: principal-model moment ratios are recoverable, and conditional on a common recovered principal part, each finite weighted correction grade is recoverable from normalized moment rates.

1 Setting

The three-tide arc for class (c) — moment ratios, the difference limit, and this recovery — follows the archived scoping consult, which also fixed what deliberately stays open: the coarea/Gelfand–Leray identification of the principal part awaits a thicker geometric measure theory in Mathlib, and the full weighted-analytic induction is a programme rather than a tide. The note’s class (c) argument is thereby formalised in its recoverable fragments, with the boundary drawn where the consult drew it.

2 The thing formalised

`mvMonomial_continuous` and `mvMonomial_zero_eq_zero`; expectation and covariance bilinearity over finite monomial combinations (`expectationUnder_combo_mul`, `covarianceUnder_combo_left`), with the combo integrabilities derived from per-pair hypotheses by factoring through $m_\alpha \cdot Q_c \cdot e^{-P}$ rather than double sums; `eq_zero_of_covariance_self_zero` (the stage C core: variance expansion under three integrabilities, `integral_eq_zero_iff_of_nonneg`, the almost-everywhere-to-everywhere step by `Continuous.ae_eq_iff_eq`, and the origin evaluation); `monomialCombo_eq_zero_of_covariance_monomials_zero`; and

`grade_eq_of_normalized_rates` (stage D), whose interface hypothesis is stage B's conclusion — callers discharge it under that theorem's explicit majorant package, keeping this file free of domination bookkeeping.

3 Where progress stalled

Two rounds, all catalogued or newly catalogued: `Finset.sum_div` wants the numerator-sum orientation (push the sum through the division first, then convert per-term with `mul_div_assoc`); the `Pi.sub` single-lambda trap fired on `integral_add` (the type-ascribed witness fixes it, as the repo playbook says); and `push_neg` is deprecated for `push Not` on this toolchain.

4 Mathlib lookup versus new mathematics

Lookup: `integral_eq_zero_iff_of_nonneg`, `Continuous.ae_eq_iff_eq`, `pow_eq_zero_iff`, `tendsto_nhds_unique`, `Finset.prod_eq_zero`, `integral_finset_sum`. New: none — variance rigidity and first-variation covariance formulas are classical; the value is the composition against an arbitrary reference density in the note's own pairwise shape.

5 What the next tide inherits

Class (c) closes here by design. What remains of the `germbij` note across all classes: the coarea/Gelfand–Leray layer (a Mathlib geometric-measure-theory prerequisite, not a tide); the concrete anisotropic instantiation of stages B–D for specific weighted families, should an application want named examples; and the two deferred nondegenerate extras (explicit Wick values for the forward coefficients; the §5 global-mixture audit). The inverse-half perimeter is closed by the parallel session's stack (PRs #107, #109, #112, #115), awaiting merge at the user's discretion.